


# Rainfall Patterns and 30-Minute Rainfall Prediction with Apache Spark

DSA5208 | Scalable Distributed Computing — Project 2

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**Abstract:** This report analyses eight years (2017–2024) of five-minute rainfall readings from Singapore's gauge network — 48.5 million records from 91 stations — using Apache Spark. Task 1 explores how rainfall varies within a year and from year to year; Task 2 builds a model that estimates the probability that a station records more than a chosen amount of rain in the next 30 minutes, taking the station, the time and the threshold as inputs. After a data-quality stage that caught two kinds of faulty gauge a range check cannot see, we built an interpretable logistic-regression baseline and then a cross-validated **random forest** on richer weather features. On the untouched 2024 test year the final model reaches a PR-AUC of **0.466** at the 5 mm threshold (56× the event rate) with ROC-AUC **0.969**, beats the logistic baseline at every threshold on identical rows, and gives calibrated probabilities. The complete notebook was also executed on a Google Cloud Dataproc cluster with identical conclusions, and doubling the cluster's workers halved the run time of the core Spark job.

# 1. Executive Summary

This project processes the data.gov.sg **Historical Rainfall across Singapore** collection (2017–2024, 7.5 GB of CSV) with Apache Spark, from raw files to a trained model, in one notebook. Spark handles every step that touches the 48.5 million readings: parsing, cleaning, the window-function features, the training sets and the MLlib models.

**Task 1** shows that Singapore's rainfall has two wet seasons, two daily rhythms, a strong west–east gradient and large year-to-year swings driven by how often it rains rather than how hard. **Task 2** delivers a single Spark MLlib model that answers P(rain in the next 30 minutes > h) for any station, time and threshold h. The final model is a **random forest** chosen by year-blocked cross-validation; on identical 2024 rows it beats the Stage 1 logistic model on PR-AUC, ROC-AUC and log loss at every threshold.

| Model | Mean PR-AUC | Mean ROC-AUC | Mean log loss | PR-AUC at 5 mm | ROC-AUC at 5 mm |
|---|---|---|---|---|---|
| Final model: random forest (Stage 2) | 0.450 | 0.973 | 0.0266 | 0.461 | 0.970 |
| Stage 1: logistic regression | 0.408 | 0.921 | 0.0328 | 0.418 | 0.912 |
| Climatology baseline | 0.023 | 0.695 | 0.0553 | 0.018 | 0.684 |

*Table 1. Final model against the Stage 1 model and the climatology baseline on identical 2024 rows (1,255,003 rows per threshold; means over the 1, 2, 5, 10 and 20 mm thresholds).*

### Key Insight

The most useful information for the next 30 minutes is **what is raining right now, nearby**: the station's own rain in the last 5–15 minutes, how long it has been dry, and the rain at gauges within 10 km. Time of day and season help, but on their own they are barely better than climatology (Model A, Section 8). A model that cannot see any rain observation is almost useless for this task; a model that can see the neighbourhood is 56 times better than chance at 5 mm.

# 2. Problem Statement & Success Metrics

**Task 1.** Explore the weather patterns in the rainfall record and how they change within a year and across 2017–2024, with analysis and visualisations.

**Task 2.** Build a model that estimates the probability that a monitoring station will accumulate more than a given amount of rain in the next 30 minutes. Inputs: station, prediction time and threshold h. The model must be built and evaluated with Spark MLlib.

## 2.1 Why accuracy is the wrong metric

Rain above 5 mm in a half-hour happens in only **0.83 %** of station-times. A model that always answers "no" is 99.16 % accurate and completely useless. We therefore judge the probabilities themselves:

| Metric | Question it answers | Good value |
|---|---|---|
| PR-AUC (average precision) | When the model flags rain, how often is it right, across all cut-offs? Compare with the event rate. | high |
| ROC-AUC | How well does it rank wet half-hours above dry ones? | high |
| Log loss | How much confident wrong probabilities are penalised (a proper scoring rule; the cross-validation metric) | low |
| Brier skill score | Share of the climatology baseline's squared error that the model removes | high |
| Reliability | Do half-hours forecast at 20 % actually turn wet 20 % of the time? | on the diagonal |
| Monotonicity | P(> 10 mm) must never exceed P(> 5 mm) for the same station and time | 0 violations |

*Table 2. Evaluation metrics used throughout. All are computed on the 2023 validation year for model choice and once on the 2024 test year.*

## 2.2 One model for every threshold

> **Design choice:** the threshold h is an **input feature** of a single binary classifier. Each training example (station, time) is paired with several thresholds, each with its own 0/1 label, so the model learns how the probability falls as h rises and can answer for any h in its trained range (0.2–60 mm). This replaces one model per threshold with one model for all.

# 3. Dataset & Computing Environment

The collection has one CSV per year with one row per station per five-minute reading. The eight files total **7.5 GB and 48.5 million rows** from 91 distinct stations; between 54–80 stations report in any one year because the network grew in 2020 and gauges come and go.

| Column group | Columns |
|---|---|
| Time | timestamp (ISO 8601, +08:00), update_timestamp, reading_update_timestamp, date |
| Station | station_id, station_name, station_device_id, location_longitude, location_latitude |
| Measurement | reading_value (mm per 5 minutes), reading_type, reading_unit |

*Table 3. Dataset schema. 12 columns, one row per station per 5-minute interval.*

## 3.1 Computing environment

The submitted run uses Apache Spark in local mode on one laptop. On **Google Cloud Dataproc**, the core Spark job was timed on 2 and 4 workers to measure horizontal scaling (Section 8.5), and the complete notebook was executed on a 2-worker cluster with Spark on YARN and the data on HDFS (Section 8.6).

| Item | Value |
|---|---|
| Machine | Apple MacBook Pro, Apple M4 Max, 16 cores (12 performance + 4 efficiency), 128 GB RAM, macOS |
| Spark | Apache Spark 3.5.5 (PySpark), master local[14], driver memory 16g, 96 shuffle partitions |
| Software | Python 3.11.17, OpenJDK 17, pandas / NumPy / Matplotlib 3.0.6 / 1.26.4 / 3.11.2 |
| Storage layers | CSV 7.5 GB → bronze Parquet 49 MB (typed raw copy) → silver 413 MB (clean readings) → gold 5.9 GB (features, training sets, predictions); Snappy-compressed, partitioned by year |
| Saved models | final random forest 11.5 MB; Stage 1 logistic baseline 13 KB |
| Cloud: scaling runs | Google Cloud Dataproc 2.2 (Debian 12, Spark 3.5), region asia-southeast1 (Singapore); master and workers n2-standard-2 (2 vCPU, 8 GB), 50 GB standard disks; 2 and then 4 workers; data in a Cloud Storage bucket; gcloud dataproc jobs submit pyspark (cloud/dataproc_benchmark.py) |
| Cloud: complete run | Dataproc 2.2, master + 2 workers n2-highmem-4 (4 vCPU, 32 GB, Intel Xeon 2.8 GHz), 100/75/75 GB balanced persistent disks (the education project's quotas: 12 vCPUs, 250 GB SSD); Spark 3.5.3 on YARN, data layers on HDFS, driver 16 GB; Cloud NAT for package installation |

*Table 4. Computing environment: the laptop as recorded by the notebook (t00_environment.csv), the cluster as configured for the Dataproc jobs.*

| Stage | Minutes |
|---|---|
| Ingestion and data-quality audit | 0.4 |
| Cleaning and pre-processing | 0.6 |
| Task 1 analyses | 0.5 |
| Stage 1 features, sets, baselines and models | 11.2 |
| Stage 2 features and feature screening | 1.2 |
| Stage 2 cross-validation (25 settings × 4 folds) | 35.2 |
| Stage 2 ablation, calibration, refit and 2024 evaluation | 25.0 |
| **Total** | **74.3** |

*Table 5. Wall-clock time of the notebook's stages on this machine in the submitted run (t40_stage_timings.csv). The laptop slept for about 10 minutes during two of the Stage 1 logistic fits (45 s and 168 s instead of the usual 10–11 s); every other stage ran uninterrupted.*

# 4. Data Cleaning

## 4.1 What the audit found

- **Values are clean:** no missing timestamps, station IDs or readings; no negative values; one reading type in one unit (mm). A single reading of 37.2 mm in five minutes, far above every other value, is an instrument fault.
- **Timestamps are not:** 1.78 million readings, all in 2017, are stamped at hh:m4:59 instead of hh:m5:00. Rounding to the 5-minute grid fixes this and reveals 86 duplicate station-slots.

- **Faulty gauges hide in plausible values:** a broken gauge reports 0.0 mm through a wet month, and one gauge reported a 749 mm "day" while its five neighbours saw at most 32 mm. Neither is caught by a range check; both are caught by comparing a gauge with its neighbours (67 dead station-months and 4 spike days).
- **What a timestamp means is undocumented.** Readings are published 2–10 minutes after their stamp, so we take a stamp as the start of its 5-minute interval. This is the conservative reading for Task 2: no future information can reach the features.

### 4.2 Cleaning steps

| Cleaning step | Records remaining | Change |
|---|---|---|
| Raw records (bronze) | 48,538,713 | — |
| After the range check (one 37.2 mm reading removed) | 48,538,712 | -1 |
| After de-duplication of station-slots | 48,538,626 | -86 |
| After removing dead-gauge station-months | 48,340,082 | -198,544 |
| After removing spike station-days (final, silver) | 48,338,938 | -1,144 |

*Table 6. Records remaining after each cleaning step. 99.59 % of the raw records are kept.*

**Note on the gauge-fault rules.** They remove only 0.4 % of the readings but change the conclusions: left in, the S107 fault would have shown up in Task 1 as a record 147 mm hour in 2022, and the dead months would pull island-wide means down. The rules are deliberately simple and documented: a station-month is dead if its rain rate is below 10 % of the network median when that median is at least 50 mm; a station-day is a spike if it exceeds 100 mm and 4× the wettest gauge within 5 km.

**Missing readings are never filled with zeros:** a missing reading is unknown, not dry. Later steps use completeness rules instead (a station-day needs 80 % of its slots, an island-day at least 20 usable gauges, and a fixed 21-gauge network checks that the changing network does not distort the results).

De-duplication keeps the most recently updated reading of each station-slot with a single max over an (update time, value) struct — one shuffle instead of a sorted window over 48 million rows. The physical plan confirms it:

```
== Physical Plan ==
HashAggregate(keys=[station_id, ts], functions=[max(struct(upd, rain_mm)), count(1),
first(station_name), ...])
+- Exchange hashpartitioning(station_id, ts, 96)
   +- HashAggregate(keys=[station_id, ts], functions=[partial_max(...), partial_count(1), ...])
      +- Project [...timestamp_seconds(round(unix_timestamp(ts_raw) / 300) * 300) AS ts...]
         +- FileScan parquet bronze/ [filtered on reading_type]
```

## 5. Weather Patterns (Task 1)

All Task 1 quantities are Spark aggregations of the cleaned readings, built from station-days with completeness rules so that the growing gauge network cannot masquerade as a change in climate.

### 5.1 Year to year: dry years have fewer rain events, not lighter rain

| Year | Rainfall (mm) | Wet days (≥ 1 mm) | mm per wet day | Events per gauge | Gauges |
|---|---|---|---|---|---|
| 2017 | 2,454 | 239 | 10.3 | 403 | 58 |
| 2018 | 2,656 | 245 | 10.9 | 374 | 56 |
| 2019 | 1,787 | 172 | 10.4 | 266 | 54 |
| 2020 | 2,631 | 230 | 11.4 | 365 | 80 |
| 2021 | 2,900 | 214 | 13.6 | 381 | 74 |
| 2022 | 2,893 | 241 | 12.0 | 437 | 72 |
| 2023 | 2,879 | 228 | 12.6 | 412 | 73 |
| 2024 | 2,708 | 233 | 11.6 | 397 | 70 |

*Table 7. Annual island-wide rainfall (gap-filled), wet days scaled to a full year, and rain events per gauge.*

Annual rainfall ranges from **1,787 mm (2019) to 2,900 mm (2021)** around a 2,613 mm mean. 2019 was dry because it rained less often — 172 wet days against 233 in the other years — while its rain per wet day (10.4 mm) was close to normal (11.8 mm). Dry and wet spells last whole seasons: the 2019 dry spell was 12 consecutive below-average months.

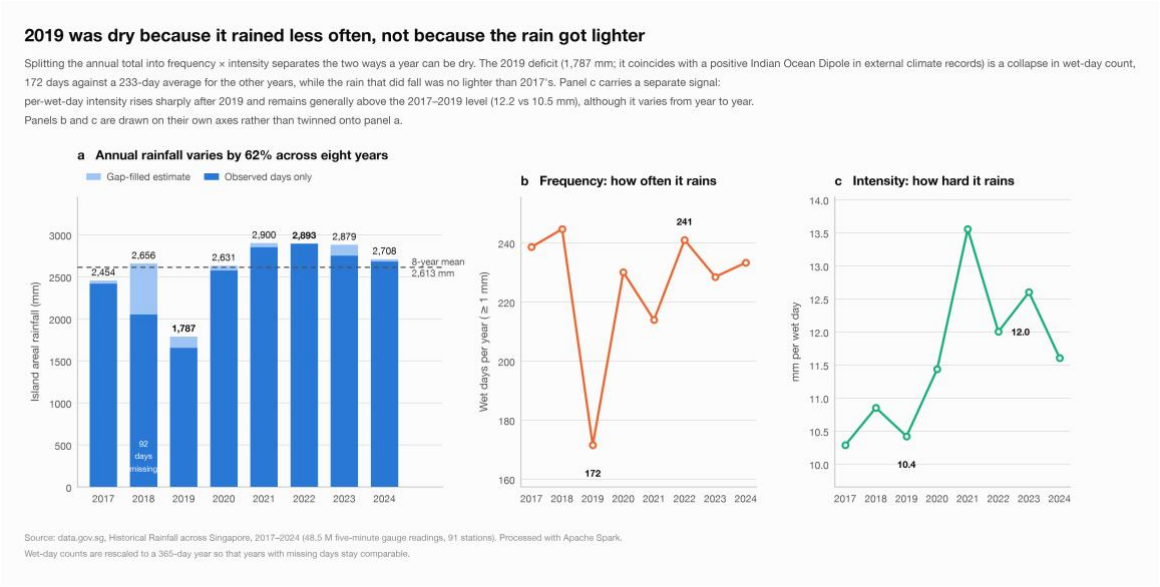



*Fig 1 – Annual rainfall (a) split into frequency (b, wet days) and intensity (c, mm per wet day).*

## 5.2 Within a year: two wet seasons and two daily rhythms

The monthly climatology has two wet seasons — **Nov (315 mm), Dec (261 mm), Apr (248 mm)** — and the driest months are **Jul (147 mm), Feb (145 mm)**. The same calendar month can differ tenfold between years, so the calendar alone predicts little. Within the day, Singapore has two rainfall clocks (Fig 2): from June to September the mean rain rate peaks at about **10:00**, consistent with squall lines arriving in the morning, while in every other season it peaks sharply at **16:00–15:00** with afternoon convection. Rain is also extremely concentrated: the wettest 10 % of wet five-minute slots deliver **49 %** of a year's rain and the wettest 1 % deliver 10 %. The typical rain event is short and light (median 20 minutes, 1.0 mm).

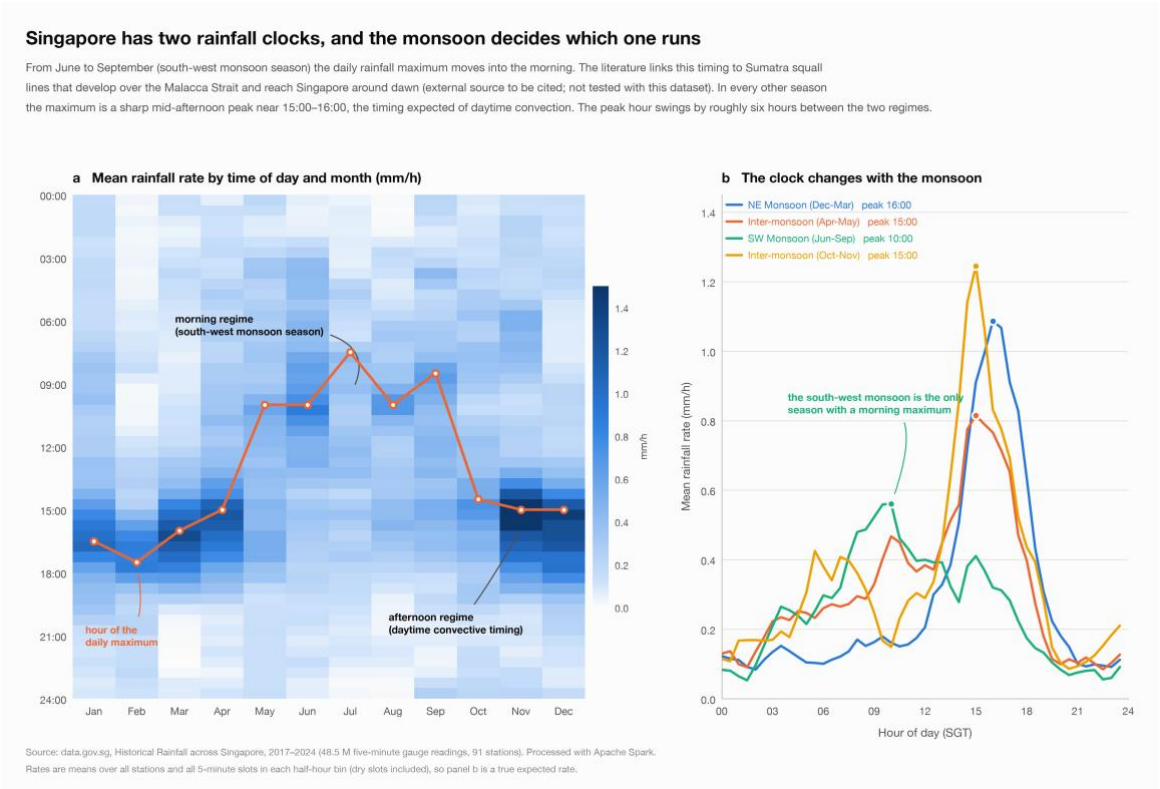



*Fig 2 – Mean rainfall rate by half-hour and month (a) and by season (b): a 10:00 peak in the south-west monsoon, a 15:00–16:00 peak otherwise.*

## 5.3 Across the island: 2× wetter in the west

Over 2021–2024, the 53 gauges with complete records average from **3,548 mm a year (S228) down to 1,700 mm (S208)**, a factor of 2.1 across an island 50 km wide; rainfall falls by about 266 mm per 10 km travelled east (r = -0.68). Rain is also local: gauges less than 2 km apart agree closely on their 30-minute totals (correlation 0.84) but beyond 20 km they are nearly independent (0.15), and the correlation drops to 1/e at about **9 km**. A single island-wide forecast cannot be right everywhere, so Task 2 predicts per station and uses neighbouring gauges within 10 km.

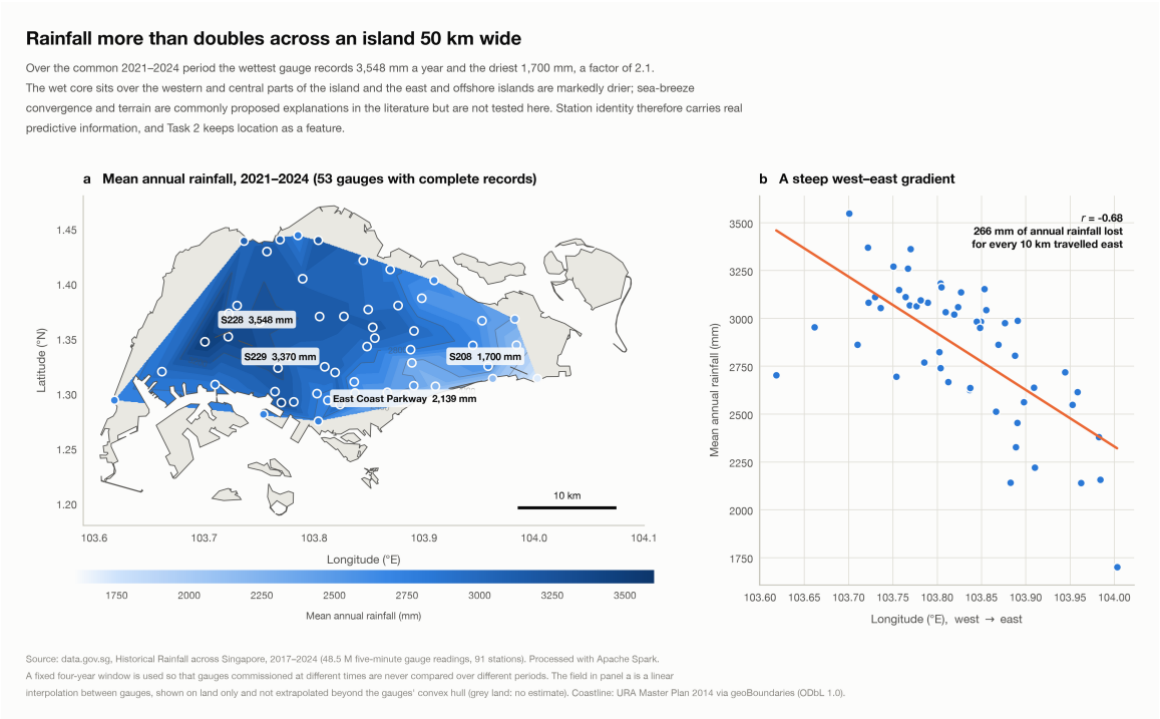



*Fig 3 – Mean annual rainfall 2021–2024 interpolated between gauges on Singapore's land area (a), and its west–east gradient (b).*

## 5.4 Heavy rain and trends

Heavy half-hours (≥ 5 mm in 30 minutes) were **+24 % more frequent in 2020–2024 than in 2017–2019** (+20 % on the fixed 21-gauge network, so the 2020 network expansion explains little of it). With only eight annual values, however, an exact Mann–Kendall test finds **no statistically significant trend in any series** (Table 8): the period is too short to separate a trend from year-to-year variability, and "not significant" means "not shown", not "no change".

| Annual series | Kendall τ | p-value | Sen slope (% of mean per year) |
|---|---|---|---|
| annual rainfall, gap-filled (mm) | +0.36 | 0.28 | +2.3 % |
| wet days per year (≥ 1 mm, scaled) | -0.07 | 0.90 | -0.3 % |
| intensity (mm per wet day) | +0.57 | 0.06 | +2.7 % |
| rain events per gauge | +0.29 | 0.40 | +1.6 % |
| 30-min totals ≥ 5 mm, rate (all gauges) | +0.43 | 0.18 | +2.4 % |
| 30-min totals ≥ 5 mm, rate (fixed network) | +0.36 | 0.28 | +1.4 % |
| annual max 30-min depth (mm, any gauge) | -0.36 | 0.28 | -1.2 % |
| annual max 60-min depth (mm, any gauge) | +0.29 | 0.40 | +1.5 % |

*Table 8. Mann–Kendall trend tests on the eight annual values (exact two-sided p-value) with Sen's slope.*

**Robustness check.** Re-deriving the annual and monthly results on the fixed network of 21 gauges that report in all eight years gives annual totals that correlate at r = 0.986 with the full network, the same driest year and wettest month, and monthly anomalies that agree in sign 97.9 % of the time.

**What Task 1 told Task 2:** use the recent rainfall state, not only the calendar (spells persist); include hour of day *and* day of year (two daily regimes switched by the monsoon); predict an exceedance probability, not an amount (heavy-tailed intensity); use station location and neighbours within 10 km (gradient and 8.5 km

correlation length); and evaluate with imbalance-aware metrics (base rate 0.83 % at 5 mm, varying from 0.13 % to 3.07 % with hour and season).

# 6. Feature Engineering

Every feature is computed with Spark window functions over each station's time series, using **time-based windows** (rangeBetween on epoch seconds) with an explicit count of the readings present, so a gap in the data never silently pulls in older readings. Neighbour features come from a small broadcast table of station pairs built from coordinates only, joined to the readings and grouped by (target station, slot), so the 48 million-row table is never joined station-by-station.

**Key design principle:** only information available at the prediction time t is used — the station's coordinates, the clock, and rain stamped at or before t − 5 min at the station and its neighbours. An independent recomputation in plain pandas of 300 random rows, from readings truncated at the anchor, matched every Spark feature and target exactly, for both stages. Neighbour denominators count only gauges that have reported *by* the prediction time (an earlier version counted gauges that reported later that day — a look-ahead that the check exposed).

| Group | Features | Why (Task 1 evidence) |
|---|---|---|
| Inputs | threshold h as log(1 + h) (Stage 1) or h and log h (Stage 2); station | the task's own inputs |
| Time | sine/cosine of time of day and day of year; season | two daily regimes, monsoon seasons |
| Location | latitude, longitude | 2× west–east gradient |
| Recent rain at the station | last 5 min; last 15/30/60/180-min totals; wet slots and peak in the last hour; **storm trend** (last 15 min minus the 15 before); **minutes since the last rain**; **readings present** in the last 1 h / 3 h; **rain so far today** | events last ~20 min; spells persist; missing data is a feature, not zero rain |
| Neighbouring gauges | Stage 1: up to 4 within 10 km (mean and max of last 30 min, mean of last 60 min, latest 5-min max). Stage 2: **three rings** — ≤ 5 km, ≤ 10 km and 5–20 km to the **west** — each with mean, max and share wet in the last 30 min, plus availability flags | correlation length ~8.5 km; squall lines arrive from the west |
| Island-wide | **share of all gauges wet** in their last 30 min and their mean rain | wet days vs dry days |

*Table 9. Feature groups. Stage 1 uses the compact set; Stage 2 adds the richer items in bold and lets cross-validation choose.*

Stage 1 keeps rows with a complete 3-hour history and all six future slots (32,962,257 training prediction times). Stage 2 keeps a row even when its history has gaps, because the number of readings present is itself a feature; it takes prediction times every 15 minutes in 2017–2023 (neighbouring 5-minute rows are near-duplicates) and every 5 minutes in the 2024 test year.

# 7. Modelling Approach

## 7.1 Chronological split and year-blocked cross-validation

**No random splits.** Consecutive five-minute examples share five of their six future slots, so a random split would leak the answer. Stage 1 trains on 2017–2022, selects on 2023 and is tested once on 2024. Stage 2 uses Spark's CrossValidator with a fold column of **whole years** (2017–18, 2019–20, 2021–22, 2023), refits on 2017–2023, and is tested once on 2024. The test year is never used for any choice.

## 7.2 Training sets

Both stages train on samples whose selection **does not depend on the label**, so the probabilities stay calibrated without re-weighting tricks. Stage 1 keeps a row for threshold h if a hash of (station, time, h) falls below a rate that depends only on h (about 8,000 positives per threshold); the positive rate of the sample equals the population rate at every threshold. Stage 2 keeps every row with rain in the last 3 hours or the next 30 minutes and 3 % of the quiet

rows with weight 33, which MLlib uses through weightCol; each row is paired with three thresholds drawn log-uniformly from 0.2–60 mm.

### 7.3 Models evaluated

| Candidate | PR-AUC | ROC-AUC | Log-loss skill | Threshold violations |
|---|---|---|---|---|
| Climatology (season × hour × threshold) | 0.043 | 0.720 | 0.000 | — |
| Base rate per threshold | 0.018 | 0.500 | -0.059 | — |
| Logistic: own gauge only | 0.403 | 0.899 | 0.325 | 0.00 % |
| Logistic: + 4 neighbouring gauges | 0.445 | 0.929 | 0.373 | 0.00 % |
| Logistic: + threshold interactions | 0.458 | 0.929 | 0.393 | 0.00 % |
| Logistic: + interactions, stronger L2 | 0.463 | 0.930 | 0.383 | 0.00 % |
| GBT, depth 5 (80 trees) | 0.452 | 0.925 | 0.029 | 0.03 % |
| GBT, depth 7 (80 trees) | 0.449 | 0.933 | 0.031 | 0.22 % |

*Table 10. Stage 1 candidates: Spark ML pipelines (StringIndexer → OneHotEncoder → VectorAssembler → classifier) fitted on 2017–2022 and scored on 2023, macro-averaged over six thresholds. Selection rule, fixed in advance: highest log-loss skill.*

Each step of the logistic model adds skill: neighbouring gauges lift PR-AUC from 0.40 to 0.44 and the threshold interactions lift log-loss skill to the best of all candidates. The Stage 1 GBTs rank about as well but their probabilities are poor (log-loss skill 0.03) because Spark's GBT starts every prediction at 0.5 and moves by about the step size per tree, so 80 trees cannot reach the 0.1 % probabilities the 20 mm threshold needs — the reason Stage 2 adds a calibrator.

| Model family | Settings tried | Best setting | CV log loss (mean ± sd over folds) |
|---|---|---|---|
| Logistic regression | 6 | regParam=0.001, elasticNetParam=0 | 0.0392 ± 0.0021 |
| Random forest | 4 | numTrees=100, maxDepth=12 | 0.0320 ± 0.0018 |
| Gradient-boosted trees | 12 | maxDepth=4, maxIter=100, stepSize=0.1 | 0.0404 ± 0.0017 |

*Table 11. Stage 2: best cross-validated setting per model family (4 year-blocked folds, weighted log loss, 22 settings in total). The selector size k was tuned first with logistic regression (k=33: 0.0425, k=16: 0.0450, k=8: 0.0585); all 33 candidates were kept.*

### 7.4 Calibration and consistency

Tree probabilities are not automatically calibrated. A Platt-style calibrator (an MLlib logistic regression on logit p, log h and their product) is fitted on the first half of 2023 and judged on the second half; it is kept only if it lowers the held-out log loss. For the random forest it was kept (held-out log loss 0.0298 → 0.0292); for the GBT it reduced the held-out log loss from 0.0385 to 0.0315. Finally, because a tree model does not guarantee P(> 10 mm) ≤ P(> 5 mm), the probabilities of each station-time are made non-increasing across thresholds with a running minimum, which changed at most 0.062 % of the 2024 predictions. The Stage 1 logistic model is monotone by construction (0 violations in 25,771,260 adjacent pairs).

## 8. Results

### 8.1 The final random forest on every 2024 row

The final model was refitted on 2017–2023 and scored on **every station and five-minute time of 2024** (6,277,148 prediction times × six thresholds), with no sampling and no weights. Precision and recall use the F1-optimal cut-off chosen on 2023 and applied unchanged.

| h (mm) | Event rate | PR-AUC | ROC-AUC | Log loss | Brier skill | Cut-off | Precision | Recall | F1 |
|---|---|---|---|---|---|---|---|---|---|
| 0.5 | 3.63 % | 0.721 | 0.964 | 0.0645 | 0.492 | 0.40 | 0.71 | 0.59 | 0.65 |
| 1 | 2.57 % | 0.667 | 0.965 | 0.0516 | 0.448 | 0.35 | 0.65 | 0.58 | 0.61 |
| 2 | 1.69 % | 0.585 | 0.965 | 0.0397 | 0.383 | 0.25 | 0.59 | 0.52 | 0.55 |
| 5 | 0.84 % | 0.466 | 0.969 | 0.0239 | 0.295 | 0.20 | 0.49 | 0.46 | 0.47 |
| 10 | 0.38 % | 0.349 | 0.973 | 0.0129 | 0.212 | 0.15 | 0.41 | 0.39 | 0.40 |
| 20 | 0.10 % | 0.186 | 0.981 | 0.0046 | 0.098 | 0.10 | 0.28 | 0.28 | 0.28 |

*Table 12. Final model on the 2024 test year by threshold. Brier skill = share of the climatology error removed.*

At the 5 mm threshold (event rate 0.84 %) the model reaches **PR-AUC 0.466**, 56 times the event rate, with ROC-AUC 0.969, and removes 30 % of the climatology error; at its 2023 cut-off it catches 46 % of the events with 49 % precision. Skill falls as events get rarer (PR-AUC 0.721 at 0.5 mm, 0.186 at 20 mm) while ROC-AUC stays above 0.96 throughout. The probabilities are reliable where it matters: below 10 %, where almost all forecasts lie, the observed frequency matches the forecast within 0.1 points, and the expected calibration error at 5 mm is 0.0014.

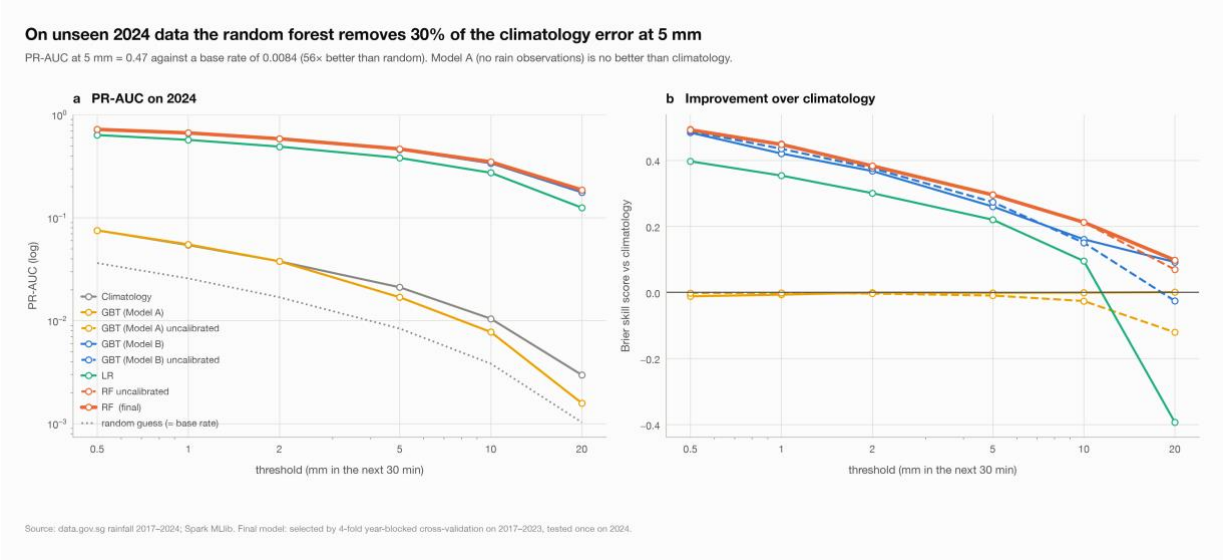



*Fig 4 – PR-AUC (a) and Brier skill against climatology (b) on all of 2024 for every Stage 2 model. Model A, which sees no rain observation, is no better than climatology.*

## 8.2 Model families at 5 mm

| Model | PR-AUC | ROC-AUC | Log loss | Brier skill |
|---|---|---|---|---|
| Climatology (station × month × hour) | 0.021 | 0.679 | 0.0483 | 0.000 |
| GBT, Model A: threshold + time + station only | 0.017 | 0.684 | 0.0472 | -0.001 |
| Logistic regression (tuned) | 0.381 | 0.951 | 0.0289 | 0.220 |
| GBT, Model B (tuned, calibrated) | 0.459 | 0.966 | 0.0262 | 0.260 |
| Random forest, uncalibrated | 0.466 | 0.970 | 0.0238 | 0.296 |
| Random forest, calibrated | 0.466 | 0.969 | 0.0239 | 0.295 |

*Table 13. Stage 2 models on all 2024 rows at the 5 mm threshold. Model A has only the threshold, time and station.*

## 8.3 Head-to-head on identical rows

The two stages evaluate different row sets, so for a like-for-like comparison both models were scored on the 1,255,003 Stage 1 evaluation rows per threshold, matched on station, time and threshold; the matched labels agree exactly. The final model wins on PR-AUC at 5 of 5 thresholds, on ROC-AUC at 5 of 5 and on log loss at 5 of 5. The largest gain is in ranking: knowing how long it has been dry, whether it is raining elsewhere on the island and whether a storm is building tells the model in advance which dry half-hours will turn wet.

| h (mm) | Events | PR-AUC clim. | PR-AUC Stage 1 | PR-AUC final | ROC-AUC Stage 1 | ROC-AUC final | Log loss Stage 1 | Log loss final |
|---|---|---|---|---|---|---|---|---|
| 1 | 32,509 | 0.049 | 0.585 | 0.663 | 0.918 | 0.967 | 0.0661 | 0.0521 |
| 2 | 21,290 | 0.033 | 0.524 | 0.585 | 0.915 | 0.968 | 0.0489 | 0.0397 |
| 5 | 10,445 | 0.018 | 0.418 | 0.461 | 0.912 | 0.970 | 0.0287 | 0.0237 |
| 10 | 4,790 | 0.010 | 0.327 | 0.350 | 0.917 | 0.975 | 0.0150 | 0.0129 |
| 20 | 1,323 | 0.004 | 0.186 | 0.191 | 0.941 | 0.982 | 0.0050 | 0.0046 |

*Table 14. Stage 1 logistic model vs the final model on identical 2024 rows.*

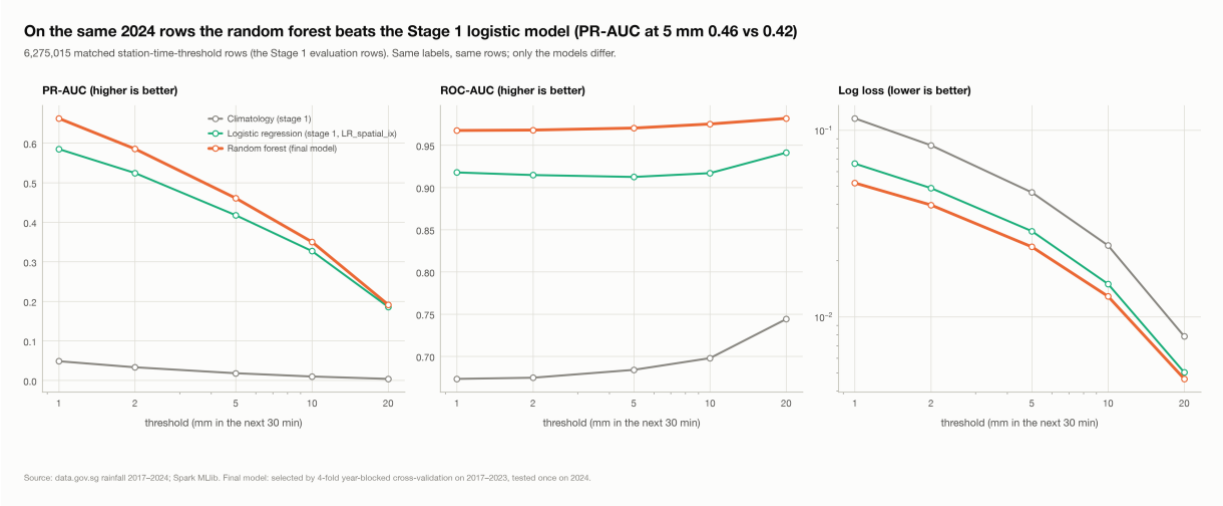



*Fig 5 – PR-AUC, ROC-AUC and log loss by threshold for the climatology, the Stage 1 logistic model and the final model on identical 2024 rows.*

## 8.4 Scalability: the same Spark job on 2 to 16 cores

To see how the work divides across cores, the heart of the cleaning stage — read the 48.5 million bronze rows, snap the timestamps, de-duplicate (one shuffle), aggregate to station-days (a second shuffle) — was run in a fresh Spark session with local[2], local[4], local[8] and local[16]. 16 cores run it **4.5× faster than 2** (26 s → 6 s), a parallel efficiency of 56 %. The gain flattens at high core counts because the two shuffles and the single SSD are shared, and because 4 of the 16 cores are efficiency cores — the usual Amdahl picture, and the reason a cluster with its own disks per node scales better than one machine.

| Cores | Wall time (s) | Speed-up vs 2 cores | Ideal | Parallel efficiency |
|---|---|---|---|---|
| 2 | 26.3 | 1.00× | 1× | 100 % |
| 4 | 13.4 | 1.96× | 2× | 98 % |
| 8 | 8.0 | 3.29× | 4× | 82 % |
| 16 | 5.9 | 4.46× | 8× | 56 % |

*Table 15. Wall time and speed-up of the cleaning job against the number of cores (t42_scaling.csv).*

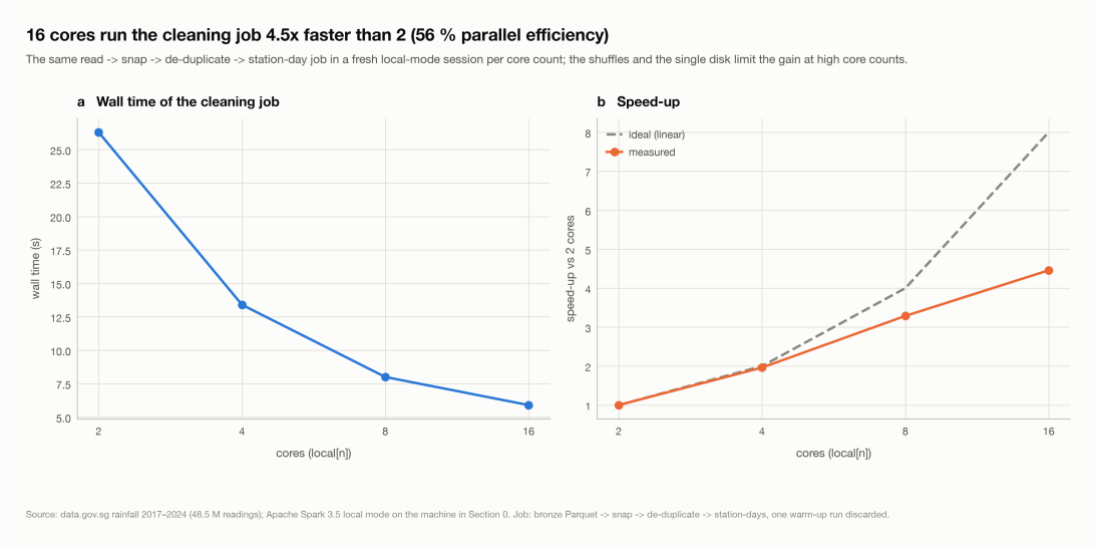



*Fig 6 – Wall time (a) and speed-up (b) of the cleaning job on 2, 4, 8 and 16 cores.*

## 8.5 The same job on a Google Cloud Dataproc cluster

The identical job was submitted to a Dataproc cluster with 2 and then 4 workers (n2-standard-2), reading the bronze layer from Cloud Storage; it returned exactly the same answer as the laptop. Ingesting the 7.5 GB of CSV from the bucket into bronze Parquet took 328 s on 2 workers. For the cleaning job, **doubling the workers cut the time from 74.8 s to 37.3 s, a 2.0× speed-up**, whereas on the laptop going from 4 to 8 cores gave 1.7× (13.4 s → 8.0 s).

| Platform | Configuration | Cores | Wall time (s) | Speed-up | Efficiency |
|---|---|---|---|---|---|
| Laptop | local[2] | 2 | 26.3 | 1.00× | 100 % |
| Laptop | local[4] | 4 | 13.4 | 1.96× | 98 % |
| Laptop | local[8] | 8 | 8.0 | 3.29× | 82 % |
| Laptop | local[16] | 16 | 5.9 | 4.46× | 56 % |
| Dataproc cluster | 2 workers x n2-standard-2 | 4 | 74.8 (73.4-76.1) | 1.00× | 100 % |
| Dataproc cluster | 4 workers x n2-standard-2 | 8 | 37.3 (33.9-43.7) | 2.01× | 100 % |

*Table 16. The cleaning job on the laptop and on the Dataproc cluster (cluster: mean of three timed runs, range in brackets; t44_local_vs_cloud.csv).*

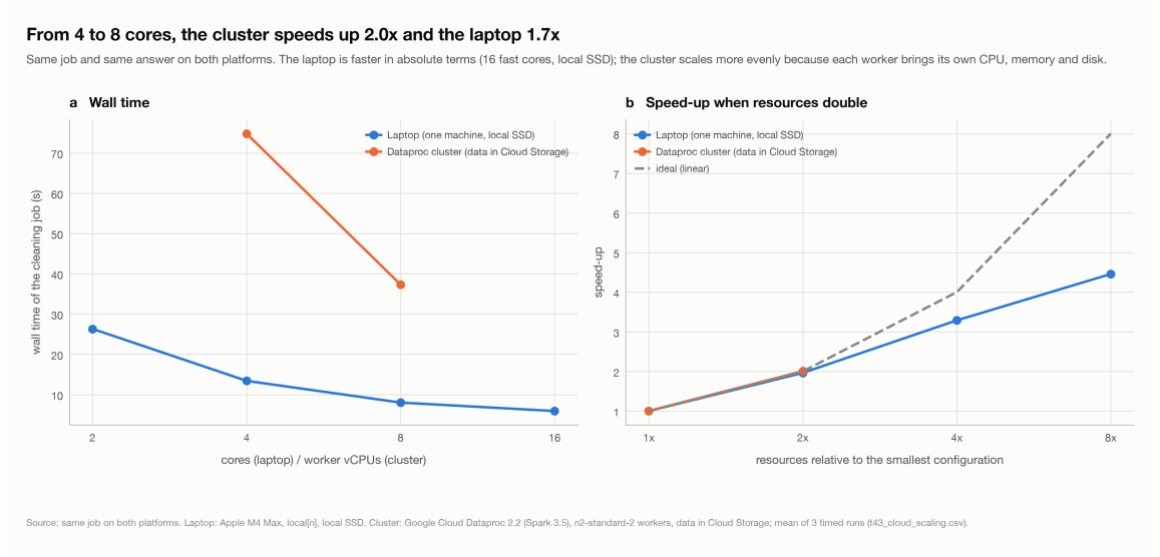



*Fig 7 – Wall time (a) and speed-up when resources double (b): laptop cores against Dataproc workers for the same job.*

**What this shows.** The laptop is about 5× faster than the cluster at the same core count: it has fast cores, 128 GB of memory and a local SSD, while each n2-standard-2 worker has 2 vCPUs, reads its input from Cloud Storage over the network (each request took 0.3–0.5 s in the job logs) and spills shuffles to standard disks. But adding cores to one machine shares one memory bus and one disk, so its efficiency falls (82 % at 8 cores, 56 % at 16), while adding workers adds CPU, memory and disk together, so the cluster scaled almost perfectly from 2 to 4 workers.

**Takeaway:** at 7.5 GB a single strong machine is the fastest and cheapest option; a cluster pays off once the data or the model no longer fits on one machine, and its near-linear scaling is what lets it grow. The education project's 12-vCPU quota limited the experiment to 4 workers.

## 8.6 The complete pipeline on the Dataproc cluster

Finally the **whole notebook** — cleaning, Task 1, both stages of Task 2 with the full cross-validation, the 2024 evaluation and predict() — was executed on a Dataproc cluster: a master and 2 workers n2-highmem-4 (4 vCPU, 32 GB; the 12-vCPU quota), Spark 3.5.3 on YARN, the data layers on HDFS. The code is the submitted notebook with only the Spark master, the storage and the laptop core benchmark changed (cloud/make_dataproc_notebook.py); the executed copy is in cloud/full_run/. It ran without errors in 290 minutes of timed work against 74 on the laptop, 3.9× slower, for the same reasons as in Section 8.5: 8 executor cores on 2.8 GHz Xeons whose shuffles cross the network, against 14 fast cores that share one memory.

| Stage | Laptop (min) | Cluster (min) | Cluster / laptop |
|---|---|---|---|
| Ingestion and data-quality audit | 0.4 | 2.6 | 6.1× |
| Cleaning and pre-processing | 0.6 | 2.5 | 4.0× |
| Task 1 analyses | 0.5 | 2.5 | 4.8× |
| Stage 1: features, sets, baselines and models | 11.2 | 30.2 | 2.7× |
| Stage 2: features and screening | 1.2 | 5.1 | 4.1× |
| Stage 2: cross-validation (100 fits) | 35.2 | 161.9 | 4.6× |

| Stage | Laptop (min) | Cluster (min) | Cluster / laptop |
|---|---|---|---|
| Stage 2: ablation, calibration, refit, 2024 test, head-to-head | 25.0 | 84.8 | 3.4× |
| **Total** | **74** | **290** | **3.9×** |

*Table 17. The same notebook on the laptop and on the Dataproc cluster: minutes per stage (t40_stage_timings.csv of each run).*

| Result | Laptop | Cluster |
|---|---|---|
| Cleaning, Task 1 tables and Stage 1 data sets (9 tables) | identical | identical (largest difference 1e-12) |
| Stage 1 logistic model: 2024 macro PR-AUC | 0.4462 | 0.4462 |
| Cross-validation: best setting of all four grids | — | same settings |
| Cross-validation: random-forest log loss | 0.0320 | 0.0319 |
| Final random forest on 2024: PR-AUC at 1 / 5 / 20 mm | 0.667 / 0.466 / 0.186 | 0.666 / 0.465 / 0.189 |
| Head-to-head at 5 mm: final vs Stage 1 PR-AUC | 0.461 vs 0.418 | 0.460 vs 0.418 |

*Table 18. Results of the two runs. Everything deterministic is identical; the tree models differ only in the third decimal because their row subsampling depends on partitioning.*

The only Task 1 value that differs is the 99th-percentile event depth (by 0.26 mm), computed with Spark's approximate percentile, whose answer depends on how the data are partitioned. The cluster run therefore confirms that the pipeline is portable: the same code gives the same conclusions on one machine and on a YARN cluster.

### 8.7 Verification

- **Metrics cross-checked against MLlib:** at 5 mm on 2023 the binned average precision (0.4448) and ROC-AUC (0.9268) agree with BinaryClassificationEvaluator (0.4417, 0.9269); the Stage 2 metrics use the evaluator directly.
- **Leakage checks passed:** targets and features recomputed in pandas from truncated data match Spark for both stages; no anchor with a missing future slot entered the feature table.
- **Reproducible:** the pipeline reproduces the group's earlier script runs exactly for Task 1 and the logistic model; Spark's tree models shift in the third decimal between runs because their row subsampling depends on data partitioning.

## 9. Interpretation & Insights

### 9.1 What drives the forecast

| Feature | Group | Importance | Plain-English explanation |
|---|---|---|---|
| thr | mandatory input | 0.191 | The threshold itself: the probability falls steeply as the asked-for amount rises. |
| log_thr | mandatory input | 0.166 | Log of the threshold, the same input on a scale the trees split on more easily. |
| r5 | recent rain | 0.149 | Rain at the station in the last 5 minutes. Rain now is the best predictor of rain in the next half-hour. |
| r15 | recent rain | 0.073 | Rain in the last 15 minutes: is it a passing shower or a storm? |
| outer_mean | neighbours | 0.069 | Mean 30-minute rain at gauges within 10 km: is a storm nearby? |
| min_since_rain | recent rain | 0.055 | Minutes since the gauge last saw rain (capped at 24 h). A long dry spell makes rain in the next 30 minutes unlikely. |
| r30 | recent rain | 0.041 | Rain in the last 30 minutes. |
| inner_mean | neighbours | 0.036 | Mean 30-minute rain at gauges within 5 km. |
| outer_max | neighbours | 0.034 | The wettest gauge within 10 km. |
| inner_max | neighbours | 0.032 | The wettest gauge within 5 km. |

*Table 19. Most important features of the final model (mean decrease in impurity) with a plain-English reading.*

The feature-group ablation agrees (GBT retrained without each group, validated on 2023): removing the station's **recent-rain** features raises the log loss by +8.0 %, removing the **neighbours** by +2.5 % and the **island-wide** signal by +0.8 %, whereas removing time costs +0.3 % and removing latitude/longitude nothing (the station index already carries location). Model A, with no rain observation at all, is +86.6 % worse.

### 9.2 Where the model works and where it struggles

- **Best in the Inter-monsoon (Oct-Nov)** (Brier skill 0.33 at 5 mm) and weakest in the NE Monsoon (Dec-Mar) (0.27): large, organised monsoon rain is well signalled by neighbours, scattered convective showers less so.
- **Weakest around 12:00** (Brier skill 0.11) and strongest at 23:00 (0.49): afternoon convective storms often start with little warning in the gauge record, whereas night-time rain is usually part of a system already visible at other gauges.
- **Heavy thresholds are hardest:** at 20 mm (0.1 % of half-hours) PR-AUC is 0.186; ROC-AUC stays high, so the model still ranks the risky half-hours well, but it cannot say *how* heavy a storm will be.
- **A single forecast can miss.** A storm can stop or start without warning; the probabilities are judged over many forecasts, which is what the reliability diagrams do.

### 9.3 How Spark was used to make this scale

- **Parse once, read many times:** the CSVs are read once with an explicit schema into year-partitioned Parquet; every later step reads the compact columnar copy (49 MB instead of 7.5 GB).
- **One shuffle where a sort would do two:** de-duplication with a struct-max aggregate; station-days and island-days as grouped aggregates; time-range window functions for every rolling feature.
- **Broadcast the small side:** the station-pair tables (neighbours, rings, spike checks) are broadcast and joined to the readings, so the large table is shuffled only by (station, slot).
- **Cache what is reused, release what is not:** the de-duplicated table, the silver readings and the training sample are persisted while several steps read them and released before the next stage.
- **Deterministic sampling:** hash-based sampling gives identical training and evaluation sets on every run and on any cluster, without a global sort.

## 10. Limitations, Risks & Future Work

| Limitation | Detail |
|---|---|
| One test year | 2024 is the only untouched year; eight years are too short to separate climate trends from variability. |
| Rare heavy rain | Only a few hundred training positives above 30 mm, so probabilities for very heavy thresholds are the least certain. |
| Gauge data only | The model sees only rain already falling on a gauge. Radar, satellite or wind fields would show storms before they reach any gauge. |
| Grid edges | Some best settings (RF trees and depth, GBT depth and iterations) lie on the edge of their search grid; a wider grid could improve them further. |
| Cluster size | The 12-vCPU quota limited the cluster to 2–4 small workers, so the cluster ran slower than the laptop; the submitted model is the laptop-trained one (the complete cluster run reproduced its results). |
| Timestamp meaning | The data documentation does not say whether a stamp marks the start or end of its interval; we chose the reading under which no future information can leak, at worst labelling the prediction time five minutes late. |

*Table 20. Data and model limitations.*

**Future work:** add radar or satellite inputs; tune beyond the grid edges; run the model training on a larger Dataproc cluster with local SSDs; and recalibrate the model each season as the gauge network changes.

## 11. Conclusion

Starting from 7.5 GB of raw five-minute readings, a disciplined Spark pipeline — explicit schema, Parquet layers, neighbour-based gauge-fault detection and completeness rules — produced a clean, trustworthy record of Singapore's rainfall. Task 1 showed two wet seasons, two daily rhythms, a 2× west–east gradient and dry years caused by fewer events. Task 2 turned those findings into features and delivered a single Spark MLlib model for any station, time and threshold: a random forest chosen by year-blocked cross-validation that reaches a PR-AUC of 0.466 at 5 mm on the untouched 2024 year, beats the interpretable logistic baseline at every threshold on identical rows, and gives probabilities one can act on.

The key takeaway is that **data quality and the information boundary matter more than the model family**: catching faulty gauges changed Task 1's conclusions, and the leakage checks caught a look-ahead that no metric

would have flagged. Once the features were right, the choice between logistic regression, random forest and boosting moved the result by far less than the features themselves.

**Declaration of AI usage** – Generative AI (Claude, Anthropic) was used as a coding assistant for the PySpark pipeline, the Dataproc benchmark, figures and debugging; to discuss design choices (the threshold-as-input formulation, label-independent sampling, leakage checks, evaluation metrics); to review and consolidate the group's two pipelines into one notebook; and to draft the README and this report from the pipeline's outputs. All numbers in this report are produced by the submitted notebook. The authors reviewed the code, the results and the text, and take responsibility for the submission.

**References.** Data.gov.sg, Historical Rainfall across Singapore (collection 2279). Apache Spark 3.5.5 MLlib Guide. Runfola et al. (2020), geoBoundaries, PLOS ONE 15(4) (Singapore coastline, ODbL). Mann (1945), Kendall (1975), Sen (1968) for the trend test. Brier (1950) for the Brier score.

**Submission.** Source code: DSA5208_Project2.ipynb (one notebook, run top to bottom, every output kept) with README.md and environment.yml. Trained model: models/task2_final_rf/ (Spark PipelineModel + calibrator + feature_spec.json); Stage 1 baseline in models/task2_baseline_lr/. Cloud benchmark: cloud/ (job script and gcloud commands). Figures and tables: outputs/.